

1. Gramática das linguagens:

1. $\{a^n b^n \mid n \rightarrow N\}$
R: $S \rightarrow aSb \mid \lambda$
2. $\{a^n b^k c^m \mid k = n + m\}$
R: $S \rightarrow AC, A \rightarrow aAb \mid \lambda, C \rightarrow bCc \mid \lambda$
3. $\{a^n b^k c^m \mid k = 2n + m\}$
R: $S \rightarrow AC, A \rightarrow aAb b \mid \lambda, C \rightarrow bCc \mid \lambda$
4. $\{a^m b^n c^i \mid m > n + i\}$
R: $S \rightarrow aS \mid aT, T \rightarrow aTb \mid U, U \rightarrow aUc \mid \lambda$
5. $\{w \in \{a, b\}^* \mid w \text{ não contém } aba\}$
R: $S \rightarrow aA \mid bS \mid \lambda, A \rightarrow aA \mid bB \mid \lambda, B \rightarrow bS \mid \lambda$
6. $\{w \mid w \text{ contém igual número de } a \text{ e } b\}$
R: $S \rightarrow aSb \mid bSa \mid SS \mid \lambda$
7. $\{ww^R \mid w \in \{a, b\}^*\}$
R: $S \rightarrow aSa \mid bSb \mid \lambda$

2. Linguagem das gramáticas:

1. $S \rightarrow aaSB \mid \lambda, B \rightarrow bB \mid \lambda$
R: $L = \{a^{2n} b^k \mid n \geq 0, k \geq 0\}$
2. $S \rightarrow aSbb \mid A, A \rightarrow cA \mid c$
R: $L = \{a^n c^m b^{2n} \mid n \geq 0, m \geq 1\}$
3. $S \rightarrow aS \mid bS \mid A, A \rightarrow cA \mid c \mid \lambda$
R: $L = \{a, b\}^* c^*$
4. $S \rightarrow abSdc \mid A, A \rightarrow cdAba \mid \lambda$
R: $L = \{(ab)^n (cd)^m (ba)^m (dc)^n \mid n, m \geq 0\}$
5. $S \rightarrow aA \mid \lambda, A \rightarrow bS$
R: $L = \{(ab)^n \mid n \geq 0\}$
6. $P \rightarrow Pc \mid Xc, X \rightarrow XbC \mid Y, Y \rightarrow aYbC \mid \lambda, Cb \rightarrow bC, Cc \rightarrow cc$
R: $L = \{a^n b^m c^r \mid m \geq n, r \geq m\}$